



Network centrality and firm performance: A meta-analysis

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Received: 12 September 2022 / Accepted: 18 July 2024 / Published online: 15 August 2024
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Abstract

This study offers a comprehensive view of the network centrality–firm performance relationship through a meta-analysis. Drawing on a data set of 1,699 effect sizes retrieved from 147 studies published during 2000–2022, the authors establish a positive association between degree, closeness, betweenness, and eigenvector measures of centrality with firm performance on average. Nevertheless, these measures show significant differences in their effectiveness across various contexts. While the associations of degree, closeness, and betweenness centralities with firm performance have diminished over time, the relationship between eigenvector centrality and performance has strengthened. Moreover, the linkage between degree centrality and overall performance is more pronounced in customer-oriented and larger markets, whereas closeness centrality demonstrates a stronger relationship with overall performance in larger markets. Also, social trust amplifies the relationships of degree, closeness, and betweenness centralities with overall performance. Furthermore, the centrality–performance linkage and the moderating effects of the contingency factors depend at least partially on the type of network (i.e., board interlock vs. alliance vs. supplier–customer) and the measure of firm performance used (i.e., innovation vs. financial outcomes). Incorporating these factors into social network analyses helps managers refine their networking strategies and enables scholars to improve the generalizability of their findings.

Keywords Social networks · Network centrality · Firm performance · Meta-analysis

“Preparation for uncertainty has become an industry norm, with executives expecting the impact of disruption—whether from geopolitical tensions, climate change effects, technology breakthroughs, or supply chain vulnerabilities—to increase by 15 to 25 percent over the next five years” (McKinsey & Company, 2024, p. 2). The evolving competitive landscape has increased the importance of fostering

marketing ecosystems that encourage collaboration and knowledge sharing to attain marketing agility in response to market dynamism (Homburg et al., 2020; Kalaignanam et al., 2021). Moreover, emerging technologies such as artificial intelligence (AI) fundamentally reshape industries, foster increasingly interconnected operations, and transform how organizations share and comprehend information (Deloitte, 2019). Further, given the increasing specialization and fragmentation across markets, firms actively seek partners with complementary capabilities to sustain a competitive edge and ensure growth potential (McKinsey & Company, 2019). For example, in response to diminishing innovation rates and rising R&D expenses, Procter & Gamble initiated its “Connect + Develop” program to foster collaboration with external partners, leveraging their expertise and resources in product development (Fang et al., 2016). Due to these dynamics, “firms and their marketing activities are increasingly nested and executed within networks of channel partners, alliances, and stakeholder arrangements” (Moorman & Day, 2016, p. 27). Such networks link firms through ties that foster collaboration and facilitate the exchange of information, knowledge, and resources (Thomaz & Swaminathan, 2015).

Stefan Wuyts served as Area Editor for this article.

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Table 1 Conceptualization of network centrality in prior literature

Network Construct	Definition	Positional Properties	Operationalization	Graphical Representation ^a	Illustrative Examples of Observed Performance Effect of Network Centrality in the Literature		
					Positive Effect	Insignificant Effect	Negative Effect
Degree Centrality	Extent to which a node interacts with other nodes	Accessing a greater volume of information faster, more efficiently, and at a lower cost (Beckman and Haunschild 2002)	$\sum_j x_{ij}$, where x_{ij} takes a value of 1 if the focal node i is connected to node j , and 0 otherwise (Freeman 1979).		Rojas, Solis, and Zhu (2018)	Swaminathan and Moorman (2009)	Shi, Yang, and Li (2019); Tao et al. (2019)
Closeness Centrality	Degree to which a node is close to other nodes	Receiving or spreading information more quickly by maintaining short distances to other nodes (Lin et al. 2009)	$1/\sum_{j \neq i} d(x_i, x_j)$, where $d(x_i, x_j)$ denotes the distance between nodes i and j (Freeman 1979)		Mani and Luo (2014)		Tao et al. (2019)
Betweenness Centrality	Degree to which a node is positioned on the shortest path between pairs of other nodes	Exerting control over information flow (Fang et al. 2016)	$\frac{\sum_j \sum_k M_{jk}(x_i)}{\sum_j \sum_k M_{jk}}$, where $M_{jk}(x_i)$ is the number of shortest paths between nodes j and k that pass through the focal node i , and M_{jk} is the number of shortest paths between j and k (Fang et al. 2016)		Egginton and McCumber (2019);		Ngataki Andersen and Gilbert (2016); Kao et al. (2017)
Eigenvector Centrality	Extent to which a focal node is close to other dominant nodes	Greater access to information that is derived from ties to powerful others (Stephen and Toubia 2010)	$\frac{1}{k} \sum_j (m_{ij} \sum_k x_{jk})$, where m_{ij} assumes a value of 1 if node i is connected to node j , and 0 otherwise; $\sum_k x_{jk}$ is the degree centrality of actor j , and k denotes the number of nodes that are linked to the focal node i (Bolander et al. 2015)		Wang, Chen, and Fang. (2018)	Watts and Koput (2019)	Kao et al. (2019)

^aEach graph represents a separate network. In each graph, ■ (resp., ▲) signifies a node that is high (resp., low) on the corresponding network construct

The significance of organizational networks is likewise evident in academic research, which acknowledges that “not only does the network or structural perspective add explanatory power to scholarly understanding of organizations’ behavior and outcomes, but ... it expands the universe of observed phenomena from an autonomous to a relational view for studying and explaining organizational action and outcomes” (Zaheer et al., 2010, p. 62). As such, existing marketing literature has extensively applied social network theory to gain a deeper understanding of interfirm relationships (e.g., Fang et al., 2016; Srinivasan et al., 2018; Thomaz & Swaminathan, 2015). Organizational network studies generally adopt a structural perspective to investigate how network position drives firm performance. In turn, they often use *network centrality*, or the extent to which a firm is connected with other network members (Freeman, 1978), to determine how organizations access and utilize information and resources.

Yet, among the many studies of organizational networks, several unresolved issues persist regarding the performance effects of network centrality. First, as Table 1 indicates, previous studies operationalize network centrality using different measures, including degree, closeness, betweenness, and eigenvector centralities. Predictably, this variation in conceptualizations and measures is mirrored in divergent findings about the performance effects of a central position within a network. For example, although Shi et al. (2019) show that degree centrality deteriorates firm performance, Mani and Luo’s (2015) findings indicate that closeness centrality is positively associated with firm performance. Similarly, Egginton and McCumber (2019) state that betweenness

centrality enhances firm performance, yet Watts and Koput (2019) find no empirical support for a performance effect of eigenvector centrality. Such divergent findings in the literature highlight the need for cumulative knowledge development efforts that synthesize and integrate previous studies of the centrality–performance linkage.

Second, diverging observations about the performance implications of network centrality go beyond its measurement. For example, a wealth of research indicates that centrality enhances performance because central firms more easily and quickly access wider information pools due to their many connections (Fang et al., 2016; Vasudeva et al., 2013). Yet the relational ties that position a central firm more deeply in a network also might lead to strategic inflexibility and information overload, thereby deteriorating the firm’s absorptive capacity (Kao et al., 2019; Tao et al., 2019).

Third, prior research predominantly focuses on firm- and network-related attributes, such as network density and diversity, as moderators of the performance impacts of network centrality (e.g., Fang et al., 2016). Considering the diverse contexts within the literature on organizational networks, research is urgently needed to integrate environmental factors into social network analyses (Shipilov, 2006). For example, organizational network studies span a wide time window characterized by continuous digital transformation; however, the dynamic nature of the relationship between network centrality and performance, and whether this relationship has evolved over time, remains largely unexplored. On the one hand, technological advancements such as AI-based predictive analytics have empowered central firms to enhance their informational advantage by adeptly integrating

and extracting insights from their broader information pool. On the other hand, recent technological advancements may have enabled even peripheral firms to collect and distribute information and knowledge through synchronous communications, potentially eroding the competitive advantage of central firms in swiftly accessing extensive volumes of information.

Similarly, market-level characteristics can significantly influence the performance effects of network centrality. For instance, holding a central position may be more advantageous in customer-oriented markets, where accessing timely information about market dynamics and customers' ever-changing demands is crucial for remaining competitive. Additionally, network centrality may be more critical in larger markets where information is more widely dispersed. Furthermore, the degree of trust among network members can play a key role in shaping the positional advantage of central firms in accessing and disseminating sensitive information within the network. Considering variations in such boundary conditions may help explain the different performance effects of network centrality observed in the prior literature.

Against this backdrop, we conduct a meta-analysis to establish a comprehensive synthesis of studies on the performance effects of network centrality. Cooperative and supply–demand relationships are two particularly significant links in influencing firms' performance within a social network (Wei & Chen, 2019). Accordingly, this meta-analysis draws on board interlock, alliance, and supplier–customer network studies. A board interlock network captures ties between firms when a director of one firm serves on another firm's board of directors (Srinivasan et al., 2018). Stuart and Yim (2010) report that approximately 85% of U.S. public firms were interlocked with at least one other organization in 2007. A strategic alliance network refers to firms connected through overlapping partnership alliances (Novoselova, 2022). A 2019 global CEO survey by PwC revealed that 40% of executives mentioned plans to enter strategic alliances in the coming year to drive business growth.¹ A supplier–customer network represents the interconnected relationships between suppliers and their customers. Business-to-business (B2B) supplier–customer relationships constitute an interconnected network that acts as the foundational framework of modern commerce. Further, the apparent shift from globalization toward near-shoring and reshoring, driven by the rise of nationalism in Western Europe and the U.S., underscores the changing landscape of geopolitics, technology, and social norms. This evolving environment demands a new

understanding of supply networks as managers navigate the unprecedented pace and intensity of market changes.

This research effort, in turn, provides several contributions to the marketing literature. First, our meta-analysis establishes a comprehensive, data-driven assessment of the relationship between network centrality and firm performance within organizational networks. This involves distinguishing among various centrality measures, including degree, closeness, betweenness, and eigenvector centralities, to offer a more nuanced understanding of the centrality–performance linkage. The evidence we gather confirms that, on average, network centrality is positively associated with firm performance. However, various centrality measures exhibit significant differences in their association with performance. For example, degree and eigenvector centralities display a stronger relationship with overall firm performance compared to closeness and betweenness centralities. This finding warns researchers against consolidating centrality measures such as through component factor analysis to simplify the interpretation of their influence on performance.

Second, we investigate contextual factors' contingent role in determining the effectiveness of network centrality. Specifically, our study is among the first to document the dynamic nature of the centrality–performance linkage. We find that the associations of degree, closeness, and betweenness centralities with overall firm performance have become weaker over time, whereas the relationship between eigenvector centrality and overall performance has strengthened. Further, we show that whether network centrality benefits firm performance is contingent on (1) the extent of customer orientation, which denotes the degree to which firms in a country prioritize customer satisfaction and retention; (2) market size, which indicates the overall size of a country's domestic economy; and (3) the degree of social trust, which refers to the level of trust among individuals in a society. Moreover, the association of centrality with firm performance and the moderating effects of the boundary conditions depend at least partially on the network type (i.e., board interlock vs. alliance vs. supplier–customer) and firm performance measure (i.e., innovation vs. financial outcomes).

Collectively, these findings offer important implications for marketing theory and practice. From a theoretical standpoint, we demonstrate that disparate centrality measures exhibit notable differences in their associations with firm performance. Further, we uncover previously unexamined moderating factors for the centrality–performance relationship that could clarify prior diverging findings. For example, we document the temporal dynamism inherent in the network centrality–firm performance relationship. From a practical perspective, our results offer managers valuable insights into the effectiveness of their networking strategies. For instance, central firms' informational advantages in markets prioritizing customer orientation become increasingly

¹ See <https://www.pwc.com/us/en/services/consulting/deals/joint-ventures-strategic-alliances.html>.

valuable. This is because network centrality enables firms to promptly identify and fill the gaps in customers' needs and requirements, thanks to their faster access to greater market intelligence.

We organize the remainder of the paper as follows: First, we discuss the theoretical background and conceptual framework. Following that, we provide a detailed description of our data, measures, and empirical estimation strategy. Next, we present our results and discuss robustness checks. Finally, we discuss the study's theoretical and managerial implications and offer an agenda for future research.

Theoretical background

With a meta-analysis, we synthesize and assess existing empirical evidence related to the firm performance effects of network centrality—defined as the extent to which a firm is linked to other network members (Freeman, 1978). Our focus on organizational networks aligns with a call from Deighton et al. (2021) to adopt a social network perspective as an informative lens on marketing problems. Such an approach is particularly relevant in B2B marketing, given its emphasis on the social nature of interfirm relationships (Grewal & Lilien, 2012). In this pursuit, we concentrate on board interlock, alliance, and supplier–customer networks, considering their widespread presence in business practice and academic literature.²

We distinguish different centrality measures in examining the relationship between network centrality and firm performance. *Degree centrality* captures the extent of a firm's interactions with network members through direct connections (Freeman, 1978). *Closeness centrality* reflects a firm's access to other firms, achieved by maintaining short distances from them (Freeman, 1978). *Betweenness centrality* denotes the degree to which a firm is positioned on the shortest path between pairs of other firms such that it can control the flow of information in a network (Fang et al., 2016). Finally, *eigenvector centrality* refers to the extent to which a firm is close to dominant players in a network (Bonacich, 1972).

Additionally, we explore the contingent nature of the centrality–performance relationship. The competitive advantage of central firms stems from their quicker access to a broader

pool of information compared to their peripheral peers. Crucially, central firms' ability to effectively integrate this knowledge into new offerings that generate value is a key factor in driving their performance. Accordingly, we examine the moderating role of factors that may influence the accessibility or competitiveness of market intelligence for central firms, as well as their ability to leverage this intelligence. Specifically, organizational network studies span an extensive time characterized by digital transformations that have fundamentally altered the quality, accessibility, and usability of information sources. Hence, we explore the dynamic nature of the network centrality–firm performance linkage to determine whether this relationship has evolved over time.

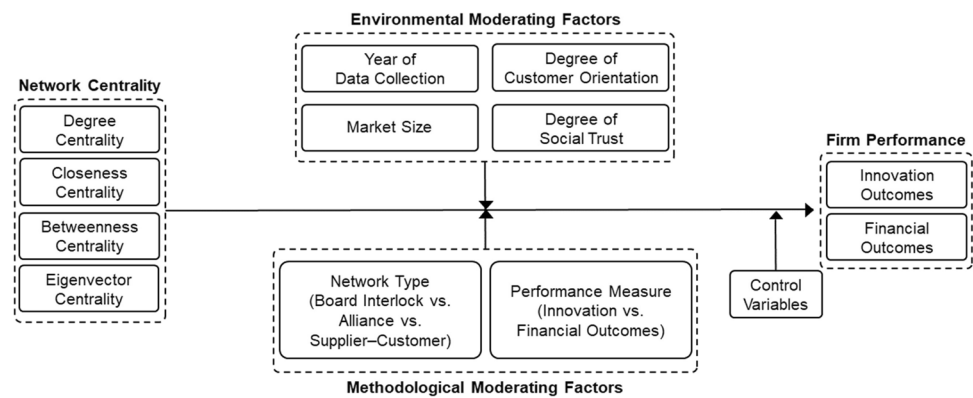
Moreover, swift access to enhanced market intelligence is particularly valuable in markets where customer satisfaction and retention are key competitive priorities. In such markets, central firms' ability to leverage their positional and informational advantages to predict future trends and understand evolving customer needs becomes an even more critical source of competitive advantage. Therefore, we investigate the moderating effect of *degree of customer orientation* (i.e., the extent to which firms operating in a country prioritize customer satisfaction and retention) on the relationship between network centrality and firm performance.

Additionally, maintaining a central position may hold greater value in larger markets, where the abundance of market participants results in more widely dispersed information (Mors, 2010). Considering their positional advantage, central firms should find integrating and distributing information in such markets easier. Therefore, we examine the moderating effect of *market size*, which reflects the overall size of a country's domestic economy, determined by its gross domestic product (GDP) and the balance of imports and exports.

Furthermore, the ability of central firms to access intelligence likely hinges on the level of trust among network members. Trust fosters a supportive environment for cooperation and communication while mitigating the risk of proprietary knowledge leaks (Briscoe & Rogan, 2016). Therefore, trust among network participants enables central firms to access a larger quantity of higher-quality information more rapidly. Accordingly, we investigate the role of *social trust* in moderating the centrality–performance relationship. Social trust reflects the extent to which individuals generally trust one another (Ma et al., 2011). At an interpersonal level, “one can examine the trust between any individuals who interact across firm boundaries...The experiences of various boundary spanners that directly interface with the partner inculcate perceptions of the partner's trustworthiness or lack thereof” (Scheer, 2022, p. 365).

Turning to a methodological perspective, substantial heterogeneity exists among previous studies regarding network types. Researchers often use social network analysis to study

² We excluded other types, such as mobility and syndication networks, for two primary reasons. First, the mechanisms through which network position can affect performance in such networks likely differ significantly from those in inter-organizational networks. Instead, we seek to establish a clear, concrete conceptual framework that specifically centers on interfirm relationships. Second, the limited number of studies of other network types prevents us from conducting meaningful meta-analyses of these research streams.

Fig. 1 Conceptual framework

the performance implications of board interlocks (e.g., Srinivasan et al., 2018), strategic alliances (e.g., Thomaz & Swaminathan, 2015), and supplier–customer relationships (e.g., Lau & Zhao, 2022). Occupying a central position within each network type may yield distinct implications for firm performance, so we examine the significance of network type (i.e., board interlock vs. alliance vs. supplier–customer) as a potential moderator of the centrality–performance relationship.

Finally, previous studies employ a diverse array of metrics to quantify firm performance, encompassing innovation-related measures such as the count of new products and patents (e.g., Srinivasan et al., 2018) and financial indicators such as sales and profit (e.g., Koka & Prescott, 2008; Thomaz & Swaminathan, 2015). However, network centrality may create dissimilar implications for different firm performance dimensions. Therefore, we explore the moderating effect of firm performance measures (i.e., innovation vs. financial outcomes). Based on these inputs, we develop the theoretically driven conceptual framework in Fig. 1.

Conceptual framework

Network centrality and firm performance

Network centrality affects firms' performance in several ways. For example, because information flows unevenly in networks, degree centrality helps firms access a wider information pool more easily (Bond et al., 2008; Fang et al., 2016). Similarly, closeness centrality makes information available at short distances, enabling central firms to enjoy faster information transfers (Tao et al., 2019). With high betweenness centrality, a firm can likely act as an information gatekeeper and control information flow across peripheral nodes (Fang et al., 2016). Also, firms that exhibit high levels of eigenvector centrality can broaden their access to information by interacting with influential partners that may be hard to reach through other channels (Bonacich & Lloyd,

2004). Such informational and positional advantages can lead to competitive advantages and enhanced performance, enabling central firms to respond to market dynamics and environmental fluctuations more quickly and effectively. For example, this acquired information can help central firms predict technological trajectories and develop new offerings that better fit customers' changing requirements (Joshi & Sharma, 2004).

Contingency factors

Year of data collection

As we detail subsequently, our meta-analysis focuses on network studies published between 2000 and 2022. Considering the increasing dynamism in business environments and rapid technological advances over this period, we explore the temporal variation in the relationship between network centrality and firm performance. On the one hand, ongoing shifts in customer preferences, intensified competitive pressures, fundamental technological disruptions, and globalization have made business environments more dynamic (Du et al., 2015). Greater market dynamism requires firms to remain agile by actively generating market intelligence about, for example, competitors' strengths and weaknesses and customers' ever-changing needs (Kohli & Jaworski, 1990).

Such demands can enhance the competitive advantages of central firms, given their prompt access to greater volumes of high-quality information. For instance, higher degree centrality can provide firms with increased access to a wealth of market intelligence, enabling them to make informed decisions in the face of increased market dynamism. Similarly, closeness centrality can enhance a firm's market responsiveness by facilitating rapid information aggregation and dissemination. Likewise, betweenness centrality can enable firms to regulate information exchange, empowering them to identify emerging trends, opportunities, and disruptions promptly. Also, enhancing eigenvector centrality through strategic connections with influential network members can

facilitate accessing timely insights and insider knowledge, thereby improving competitive responsiveness.

On the other hand, recent technological advancements empower firms to gather and disseminate information and knowledge through synchronous information sharing. This capacity can streamline information processing and retrieval efforts even for peripheral nodes (Shipilov, 2006), which might erode the competitive advantage of firms with high degree centrality in rapidly accessing larger volumes of information. Similarly, by transcending time and space, recent technological advancements may have reduced the peripheral nodes' dependence on firms with greater closeness centrality for faster information access and dissemination. By reducing informational asymmetries and narrowing the knowledge gap between network participants, peripheral nodes might gradually reduce their dependence on firms with high betweenness centrality as information intermediaries. Likewise, technological advancements may have significantly diminished the exclusivity of information once held primarily by influential network members, thereby reducing the competitive advantages associated with high eigenvector centrality. Noting these contrasting effects, the net moderating effect of the year of data collection on the network centrality–performance linkage remains an empirical question we explore.

Degree of customer orientation

We expect the degree of customer orientation in the wider economy to influence the network centrality–performance relationship. The extent of customer orientation refers to how much firms in an economy prioritize customer satisfaction and retention. Cultural and historical factors profoundly influence the degree to which businesses prioritize meeting customers' needs and requirements as a fundamental driver of success, and this emphasis can vary significantly from one country to another. In nations like Japan, characterized by deeply ingrained cultural values of respect and harmony, coupled with a strong commitment to nurturing long-term relationships and continuous improvement, there is a pronounced emphasis on meeting and surpassing customer expectations. In contrast, countries facing economic challenges and limited product supply may prioritize securing essential goods and services over catering to customers' idiosyncratic needs.

Customer-oriented economies place a greater value on market-sensing capabilities built on market information acquisition, dissemination, and interpretation (Slater & Narver, 1998). Consequently, in such markets, firms with a high degree centrality can expect their informational advantages to be more pronounced, given that their superior ability to consolidate information sources strategically positions them to enhance their market and technology sensemaking

capabilities. Similarly, firms with high closeness centrality can swiftly gather and respond to market information, enhancing their ability to adapt to changing customer demands. Also, their bridging position enables firms with high betweenness centrality to consolidate a wide range of market data and insights, providing them with a comprehensive perspective on evolving customer preferences and emerging technologies. If they have high eigenvector centrality, firms are well-connected to influential market players, allowing them to tap into a broad network of insights and expertise and stay at the forefront of industry developments. By harnessing these informational advantages, central firms can proactively predict the shifts in demand, paving the way for creating new offerings that more closely align with evolving customer preferences. These advantages will likely hold more significant value in customer-centric economies prioritizing customer satisfaction and retention.

Market size

The size of an economy can play a key role in shaping the relationship between network centrality and firm performance. Information tends to be more widely dispersed in larger economies characterized by more market participants, making knowledge integration more difficult and costlier (Mors, 2010). Holding a central position that facilitates accessing and disseminating information may become a more valuable source of competitiveness as the size of the market increases. For example, due to their many ties, firms with high degree centrality can access scattered information more easily; high closeness centrality enables the rapid spread of information among dispersed network members; betweenness centrality empowers firms to act as information gatekeepers and, hence, control the flow of scattered information; and eigenvector centrality can ensure access to critical information that may be unevenly distributed within the network.

Degree of social trust

We expect the level of social trust, defined as the extent to which individuals place trust in one another (Ma et al., 2011), to influence the relationship between network centrality and firm performance. The pressure of increased market competition, as reflected in growing R&D costs and shortened innovation cycles, prompts firms to gravitate toward collaborative partnerships (Dussauge & Garette, 1999; Hagedoorn, 1993). In this sense, building long-term relationships based on mutual trust represents an increasingly valuable source of competitiveness. The interactions of different boundary spanners who directly engage with other network members can shape their perceptions of network peers' trustworthiness (see Scheer, 2022). When trust thrives

in a society, it creates an environment in which businesses prioritize ethical conduct, nurture long-lasting collaborations, and encourage cooperation and commitment to collective interests (Briscoe & Rogan, 2016). Network actors with close, trusting relationships develop a mutual understanding of accepted norms and expected behaviors, which reduces the possibility of misinterpretation of a node's actions by other network partners (Zaheer & Bell, 2005). Moreover, strong collaborative ties that rest on trust serve as communication channels that enable exchanges of detailed, sensitive, or proprietary information (Sosa, 2011).

The presence of social trust among network participants can significantly influence the access of firms with high degree, closeness, or eigenvector centralities to a greater volume of high-quality information. Further, firms with high betweenness centrality are susceptible to exchange hazards because their bridge-like position offers arbitrage opportunities that can be used to their advantage and the detriment of peripheral firms (see Batjargal, 2010). In such a context, trust can enable firms with high betweenness centrality to foster cooperation and create environments more conducive to collaboration and knowledge sharing.

Network type

As noted, previous studies use social network analysis to investigate the effect of centrality on firm performance in board interlock, alliance, and supplier–customer networks. However, centrality within each network type may have distinct implications for firm performance. For example, centrality in board interlock networks implies access to valuable insider information and influence over corporate governance decisions. Yet central firms may benefit from broader collaboration opportunities and access to complementary resources within strategic alliance networks. Similarly, their central position in supplier–customer networks can afford firms greater control over the flow of resources in their business exchanges. To offer a more granular picture, we investigate the moderating role of network type (i.e., board interlock vs. alliance vs. supplier–customer) on the centrality–performance linkage.

Firm performance measure

Previous studies on organizational networks employ various performance measures, including innovation metrics and financial indicators. However, it is important to recognize that network centrality may have different associations with various dimensions of firm performance. The connection between centrality and innovation may be stronger than its link with financial performance. This is because not all innovations necessarily lead to economic gains, and sustaining a central position often requires significant financial

investments in relationships. Accordingly, we explore the role of the performance measure (i.e., innovation vs. financial outcomes) in moderating the centrality–performance relationship.

Other methodological characteristics

In addition to the aforementioned methodological factors, we control for several study and methodological characteristics likely to influence the observed associations between network centrality and firm performance in the existing literature. These considerations include whether partial correlations are obtained from models that control for network density and diversity, whether the primary studies use objective or subjective measures of firm performance, the data source employed to construct the network's nodes and edges (i.e., survey vs. archival data), the temporal nature of the data (i.e., longitudinal vs. cross-sectional), the method of operationalizing network centrality's effect sizes (i.e., partial or bivariate correlations), publication status (i.e., published or working paper), inclusion in top-tier academic journals featured in the *Financial Times* 50 (FT 50) list, and whether the primary study was conducted in a Western (e.g., U.S., Canada, Western Europe, Australia) or non-Western (e.g., China, India, Singapore) economic context.

Methodology

Sample

We conducted a rigorous search to identify empirical studies investigating organizational social networks. We performed an issue-by-issue review of the journals in the FT 50 list and other potentially relevant journals (e.g., *International Journal of Research in Marketing*, *Journal of Business Research*, *Industrial Marketing Management*). To identify unpublished studies (e.g., dissertations, working papers), we searched databases such as ProQuest Dissertation Express, MSI, SSRN, EconPapers, REPEC, and AISel, as well as the proceedings of prominent marketing, management, and economics research conferences. We also conducted a year-by-year search on Google Scholar. The keywords featured in the search process include “social network,” “organizational network,” “interfirm relationship,” “network centrality,” “degree centrality,” “closeness centrality,” “betweenness centrality,” “eigenvector centrality,” “board interlock,” “alliance,” “supplier network,” “customer network,” and “supply network.” We searched these keywords in articles' titles and abstracts to identify eligible studies. With a snowballing procedure, we checked the reference lists in the publications thus obtained to identify any study we might have overlooked.

Table 2 Distribution of effect sizes by publication year and country of origin

Panel A		Panel B	
Publication Year	Number of Effect Sizes	Country of Origin	Number of Effect Sizes
2000	13	Australia	59
2001	6	Canada	15
2002	14	China	270
2003	4	Finland	33
2004	11	France	42
2005	24	Germany	6
2007	50	Hing Kong	24
2008	11	India	49
2009	138	Italy	22
2010	35	Japan	9
2011	61	Mexico	21
2012	44	Norway	37
2013	51	Poland	12
2014	94	Singapore	140
2015	77	South Korea	34
2016	192	Sweden	20
2017	184	Taiwan	123
2018	129	United Kingdom	36
2019	296	United States	596
2020	69	Multi-national	73
2021	121	Unspecified country of origin	78
2022	75		

Next, we applied several decision rules to identify studies eligible for inclusion in our meta-analysis. We excluded theoretical and qualitative studies and those focused on quantitative investigations of only the antecedents of network centrality rather than its performance outcomes. Reflecting our focus on organizational networks, we also excluded studies of individual networks. We removed studies focused on other network types, such as mobility, advice, or syndication networks, to limit the sample to studies based on board interlock, alliance, and supplier–customer networks. Moreover, we excluded studies that lacked information about the bivariate and partial correlations, which we need to compute the effect sizes of network centrality on firm performance. The final sample used for path analyses comprises 1,699 effect sizes sourced from 147 studies published between 2000 and 2022. For the moderation analyses, we excluded studies with unknown or multiple countries of origin to ensure the accuracy of our measurements for the country-level moderating variables. The final sample used in the moderation analyses includes 1,548 effect sizes from 127 studies. Web Appendix A provides the list of studies in our sample; Table 2 presents the distribution of effect sizes based on publication year and country of origin.

We used the annual Global Competitiveness Reports (GCRs) published by the World Economic Forum [WEF] to obtain information about the degree of customer orientation and market size. Also, we obtained information about the degree of social trust from the Quality of Government (QoG) Institute’s database. We used a neighbor imputation method to replace the missing values for any country-level moderating variable with available values in the years before or after, depending on which one was nearer; if the values in the years before and after were equally distant from the missing value, we used their mean value.

Data coding

Employing a transparent and reproducible coding protocol equipped with comprehensive explanations for each variable, one of the authors undertook the coding of primary studies. Next, two graduate students not involved in the research project underwent thorough training and independently coded all the studies. The Cohen’s kappa coefficient for interrater reliability is 0.91, which is satisfactory (Landis & Koch, 1977). Any remaining discrepancies were resolved through discussions.

Operationalization of variables

Effect sizes

We use bivariate and partial correlations between network centrality and firm performance as proxies for the effect sizes, a common approach for meta-analytic reviews in the marketing literature (e.g., Babić Rosario et al., 2016; Eisend, 2015).³ To obtain correlation information, we first identified different network centrality and firm performance measures, as used in previous studies. Specifically, extant literature applies degree, closeness, betweenness, and eigenvector measures of centrality to determine the extent to which a firm occupies a central position in its ties with other network members (see, e.g., Fang et al., 2016; Srinivasan et al., 2018; Swaminathan & Moorman, 2009). Previous studies also use a wide range of measures to operationalize firm performance, including “innovation,” “innovativeness,” “patent count,” “patent citation,” “new product,” “new service,” “sales,” “revenue,” “profit,” “earnings,” “cash flow,” “income,” “return on assets,” “return on sales,” “return on investment,” “return on equity,” “productivity,” “stock return,” “Tobin’s q,” “stock returns,” and “risk” (see, e.g., Koka & Prescott, 2008; Srinivasan et al., 2018; Thomaz & Swaminathan, 2015). After obtaining the effect sizes, we adjusted each effect for measurement error by dividing the correlation coefficient by the product of the square root of the reliabilities of its constituent constructs (Hunter & Schmidt, 2004). We replaced any missing reliabilities for a construct with the average of the available reliability estimates in our sample.

Year of data collection

The central year of the primary study’s data collection window approximates when the study was conducted. If data about the sampling window are unavailable, which accounts for about 3% of our observations, we use the publication year. If the study is unpublished, we employ the year the primary study became available online.

Degree of customer orientation

To obtain information about the degree of customer orientation in the wider economy, we turn to the WEF’s GCRs,

which measure the extent to which firms operating in a country emphasize customer satisfaction and retention on a 7-point scale (1 = “Indifferent to customer satisfaction,” 7 = “Highly responsive to customers and seek customer retention”).

Market size

The domestic market size index provided by the WEF’s GCRs indicates the natural log of the sum of the GDP based on purchasing power parity (PPP) plus the total value (PPP estimates) of imports of goods and services minus the total value (PPP estimates) of exports of goods and services. The WEF’s GCRs indicate the normalized market size on a 7-point scale.

Degree of social trust

We obtain information about each country’s social trust level from the QoG Institute’s database. This database reports the average value of national social trust numbers from all the surveys taken in a year in a country that contain the question: “Do you believe that most people can be trusted?” The scores range from 0 (i.e., the lowest level of trust) to 100 (i.e., the highest level).

Network type

The *alliance network* dummy takes a value of 1 if the primary study is based on an alliance network and 0 otherwise. The *supplier–customer network* dummy assumes a value of 1 if the primary study is based on a supplier–customer network and 0 otherwise.

Innovation (vs. financial) performance

To account for how firm performance is measured, we use the *innovation (vs. financial) performance* dummy variable that equals 1 if the primary study deploys an innovation-related metric (e.g., number of new products or patents) to capture performance and 0 if it employs financial indicators (e.g., sales, profitability).

Control variables

We incorporate controls to determine if partial correlations originate from models controlling for network density or diversity. *Network density* is a dummy variable that takes a value of 1 if a partial correlation is derived from a model in the primary study controlling for network density and 0 otherwise. *Network diversity* is a dummy variable that takes a value of 1 if a partial correlation is obtained from a model in the primary study accounting for network diversity and 0 otherwise.

³ Similar to Babić Rosario et al. (2016) and Rosenthal (1988), we compute the partial correlation coefficient effect sizes as follows:

$$r_{xy,z} = t / \sqrt{t^2 + df}, \quad (5)$$

where t denotes the t statistic associated with the regression coefficient of the effect of network position on firm performance, and df is the degrees of freedom of the corresponding regression model.

We control for an *objective performance measure*, a dummy variable that takes a value of 1 if the primary study uses an objective measure to assess firm performance and 0 otherwise. We also control for the *data source* used to generate the network's nodes and edges in the primary study. The data source is a binary variable that takes a value of 1 if the primary study uses survey data to determine network structure and 0 otherwise. Furthermore, we control for *panel data*, a dummy variable that equals 1 if the data in the primary study have a panel structure and 0 otherwise. Then, we include the *partial correlation* binary variable as a control; it takes a value of 1 if the effect size is computed based on a partial correlation and 0 otherwise. Also, we control for *published study*, a dummy variable equal to 1 if the primary study is published and 0 otherwise. Moreover, we add a *top-tier journal* dummy, which equals 1 if the published primary study appears in one of the journals in the *FT 50* list and 0 otherwise. Finally, we control for the primary study's region of origin using a binary variable that takes a value of 1 if the study was conducted in a Western economy and 0 otherwise. Table 3 lists the variables' operationalizations; Table 4 provides the correlation coefficients for the variables.

Meta-analytic calculations

Path analyses

We use a random-effects model to summarize the performance effects of network centrality. We transformed the reliability-corrected correlations into Fisher's Z coefficients. To assign more weight to more accurate estimates with larger sample sizes, we calculated the weighted mean of Fisher's Z coefficients using the inverse of their variance (i.e., $N - 3$, where N is the sample size) as weights (Rubera & Kirca, 2012). Then, we transformed the weighted-average Fisher's Z coefficients into a standard correlational form (r) for the ease of interpretation (Babić Rosario et al., 2016).

Moderation analyses

To examine the moderating effects, we used a mixed-effects model. The retrieved effect sizes are nested within studies, which may lead to heteroskedasticity in the errors relative to traditional regression analyses (Krasnikov & Jayachandran, 2008; Rosengren et al., 2020). To account for any within-study error correlations, we ran hierarchical linear models and regressed Fisher's Z coefficients on the moderating and control variables, as follows:

$$Z_{DC_{ij}} = \alpha_{10} + \sum_k \alpha_{1k} X_{kij} + u_{1i} + e_{1ij}, \quad (1)$$

$$Z_{CC_{ij}} = \alpha_{20} + \sum_k \alpha_{2k} X_{kij} + u_{2i} + e_{2ij}, \quad (2)$$

$$Z_{BC_{ij}} = \alpha_{30} + \sum_k \alpha_{3k} X_{kij} + u_{3i} + e_{3ij}, \quad (3)$$

and

$$Z_{EC_{ij}} = \alpha_{40} + \sum_k \alpha_{4k} X_{kij} + u_{4i} + e_{4ij}, \quad (4)$$

where the subscripts i and j denote the primary study and the obtained effect size, respectively; $Z_{DC_{ij}}$ is the Fisher's Z coefficient between degree centrality and firm performance; $Z_{CC_{ij}}$ denotes the Fisher's Z coefficient between closeness centrality and firm performance; $Z_{BC_{ij}}$ is the Fisher's Z coefficient for the betweenness centrality–performance relationship; $Z_{EC_{ij}}$ reflects the Fisher's Z coefficient for the eigenvector centrality–firm performance linkage; X_{kij} denotes the moderating and control variables; α_{nk} ($n = 1-4$) includes the parameters to be estimated; u_{ni} indicates the study-level error term; and e_{nij} captures the measurement-level error term.

Results

Path analyses

We summarize the meta-analytic results for the effect of network centrality on firm performance in Table 5, Panel A. We obtained 628 effect sizes from 91 studies for the degree centrality–performance linkage. We document a significant and positive relationship between degree centrality and firm performance on average ($\bar{r}^O = 0.087$, $p < 0.01$). The 331 effect sizes that we retrieved from 49 studies for the closeness centrality–firm performance relationship reveal, on average, that the association between closeness centrality and firm performance is significant and positive ($\bar{r}^O = 0.057$, $p < 0.01$). Regarding the betweenness centrality–performance linkage, we obtained 351 effect sizes from 52 studies. We find that betweenness centrality is positively associated with firm performance on average ($\bar{r}^O = 0.050$, $p < 0.01$). For eigenvector centrality, we obtained 389 effect sizes from 57 studies. Our results reveal that eigenvector centrality is positively associated with firm performance on average ($\bar{r}^O = 0.073$, $p < 0.01$). Together, these findings provide strong evidence for the positive relationship between network centrality and firm performance. Notably, the between-group homogeneity tests using the Qb statistic unveil significant disparities among centrality measures concerning the magnitude of their associations with overall firm performance. Specifically, degree and eigenvector centralities exhibit stronger relationships with firm performance compared to closeness and betweenness centralities.

Substantial variation exists in our sample pertaining to the primary studies' network types and performance measures.

Table 3 Constructs, descriptions, and coding criteria/operationalizations

Construct	Description	Coding Criteria/Operationalization
Year of data collection	Time of data collection	The central year of each study's data collection window as an indicator of the time period when the study was conducted. If the data about the sampling window are not available, we use the publication year. If the study is unpublished, we employ the year the primary study became available online.
Degree of customer orientation	Extent to which firms prioritize customer satisfaction and retention	The extent to which firms operating in an economy place emphasis on customer satisfaction and retention on a 7-point scale (1 = "Indifferent to customer satisfaction," 7 = "Highly responsive to customers and seek customer retention"), as reported by WEF's GCRs
Market size	Overall size of the market in an economy	Sum of the gross domestic product plus the value of imports of goods and services, minus the value of exports of goods and services, as reported by the WEF's GCRs
Degree of social trust	Level of social trust in each country	The average value of national social trust numbers from all surveys taken in a year in a country that contained a question of the type "Do you believe that most people can be trusted?" The scores range from 0 (lowest level of trust) to 100 (highest level of trust), as reported by the QoG Institute
Alliance network	Whether the primary study is based on an alliance network	A dummy variable that takes a value of 1 if the primary study is based on an alliance network, and 0 otherwise
Supplier–customer network	Whether the primary study is based on a supplier–customer network	A dummy variable that takes a value of 1 if the primary study is based on a supplier–customer network, and 0 otherwise
Innovation (vs. financial) performance	Measurement of performance in the primary study	A dummy variable that equals 1 if the primary study deploys an innovation-related metric (e.g., number of new products, patents) to capture performance, but 0 if it employs financial metrics (e.g., sales, profitability)
Network density dummy	Whether a partial correlation is obtained from a model that controls for network density	A dummy variable that takes a value of 1 if a partial correlation is obtained from a model in the primary study that controls for network density, and 0 otherwise
Network diversity dummy	Whether a partial correlation is obtained from a model that controls for network diversity	A dummy variable that takes a value of 1 if a partial correlation is obtained from a model in the primary study that controls for network diversity, and 0 otherwise
Objective performance measure	Using objective measures to assess firm performance	A dummy variable that takes a value of 1 if the primary study uses an objective measure to operationalize firm performance, and 0 otherwise
Data source	Source of data used to determine the network structure	A dummy variable that takes a value of 1 if the primary study uses survey data to determine the network structure, and 0 otherwise
Panel data	Structure of data in the primary study	A dummy variable that takes a value of 1 if the data in the primary study has a panel structure, and 0 otherwise
Partial correlation	Whether an effect size is computed based on partial or bivariate correlations	A dummy variable that takes a value of 1 if an effect size is computed based on a partial correlation, and 0 otherwise
Published study	Whether the primary study is published or not	A dummy variable that takes a value of 1 if the primary study is published, and 0 otherwise
Top tier journal	An indicator of journal quality	A dummy variable that equals 1 if the primary published study appears in one of the journals in the FT50 list, and 0 otherwise
Western economy	Whether the primary study was conducted in a western economy or not	A dummy variable that takes a value of 1 if the primary study was conducted in a western economy, and 0 otherwise

Therefore, we provide the meta-analytic results for the relationships between network centrality and firm performance based on the type of network and performance measurement in Table 5, Panels B–F. We find almost consistent results for the positive association of network centrality and overall firm performance regardless of the type of network or performance measure,⁴ but the magnitude of these relationships varies across the subsamples. We explore the moderating effects of these methodological characteristics subsequently.

Moderation analyses

The results of the bivariate analyses highlight the great variation in the observed relationship of network centrality with firm performance in the primary studies. The significant values of Cochran's Q-test of homogeneity and the high values of the scale-free index of homogeneity I2 in Table 5 indicate that the observed heterogeneity in the effect sizes is greater than what is likely to result from the subject-level sampling error alone, emphasizing the need for a moderation analysis.

We provide the estimation results for the moderation analyses in Table 6; here, we focus on statistically significant effects. Panel A contains the results based on the full sample. It reveals that the associations of degree (Model 1: $\alpha = -0.006$, $p < 0.01$), closeness (Model 2: $\alpha = -0.007$, $p < 0.01$), and betweenness centralities (Model 3: $\alpha = -0.008$, $p < 0.01$) with firm performance have weakened over time, likely because the competitive advantages of firms with these centralities stem from their direct connection and proximity to other nodes in the network (Fang et al., 2016). This relative advantage has deteriorated with technological advancements that surpass the boundaries of time and space and grant even peripheral nodes instant access to market knowledge (Bloom et al., 2014).

Conversely, the coefficient for the year of data collection in Model 4 is positive and significant ($\alpha = 0.004$, $p < 0.1$), suggesting that the eigenvector centrality–performance linkage has strengthened over time. Because technological advancements make access to information more equitable across firms, including those on the periphery of networks, the critical determinant may have shifted from pure knowledge to the quality of connections and relationships that a firm maintains. Firms possessing high eigenvector centrality are more adept at leveraging these relationships due to their ability to establish strong connections with influential network members who hold elevated status within their networks.

In addition, the degree of customer orientation in the wider economy positively moderates the association of degree centrality with firm performance (Model 1: $\alpha = 0.039$, $p < 0.01$). Firms with higher degree centrality benefit from improved access to market intelligence, including insights about customers and competitors. In turn, they can identify market gaps and tailor their products and services to increase customer satisfaction and retention.

Furthermore, market size strengthens the associations of degree (Model 1: $\alpha = 0.062$, $p < 0.01$) and closeness centralities (Model 2: $\alpha = 0.020$, $p < 0.05$) with overall firm performance. Information is more dispersed and flows more unevenly in larger markets. Due to their many connections, firms with higher degree centrality can access scattered information more easily and efficiently (Bond et al., 2008; Fang et al., 2016). Likewise, because of their proximity to other nodes, firms with high closeness centrality can access and distribute sparse information more rapidly. Capitalizing on these strategic advantages in larger markets where information is more widespread can give central firms a competitive edge over peripheral peers with limited capacity for integrating and disseminating information.

Moreover, the degree of social trust positively moderates the associations of degree (Model 1: $\alpha = 0.087$, $p < 0.01$), closeness (Model 2: $\alpha = 0.064$, $p < 0.01$), and betweenness centralities (Model 3: $\alpha = 0.042$, $p < 0.05$) with firm performance. The positional advantage gained by firms with high degree or closeness centralities reflects their ability to maintain direct relationships with other nodes in the network. Trust facilitates relationship building and helps firms with high degree or closeness centralities establish more direct ties with other nodes in the network (Briscoe & Rogan, 2016). Similarly, due to their intermediary role, firms with high betweenness centrality are inevitably susceptible to exchange hazards (Batjargal, 2010). Potential exchange hazards threaten network peers' confidence in firms with high betweenness centrality and lower their willingness to share critical information and resources beyond what is required for a specific transaction (Markóczy et al., 2013). Promoting trust can help network members overcome these information exchange barriers.

In addition, our findings reveal a stronger association between degree centrality and firm performance in board interlocks compared to alliance networks (Model 1: $\alpha = -0.063$, $p < 0.01$). Board members are instrumental in shaping corporate strategies, whereas alliance networks typically revolve around specific projects or objectives. Firms with high degree centrality in board interlocks can exert substantial influence across multiple firms through their board representatives, thereby shaping industry-level corporate strategies, decisions, and governance practices on a broader scale. Conversely, eigenvector centrality holds greater significance in alliance networks than in board interlocks (Model

⁴ The associations of closeness, betweenness, and eigenvector centralities with overall firm performance within supplier–customer networks are insignificant. However, the limited number of related observations obtained from existing literature requires caution when interpreting these findings.

Table 4 Bivariate correlations for network centrality–firm performance effect sizes, moderators, and control variables

Constructs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Effect Sizes (Fisher's Z coefficients)																			
1 Degree centrality–Firm performance																			
2 Closeness centrality–Firm performance	–																		
3 Betweenness centrality–Firm performance	–	–																	
4 Eigenvector centrality–Firm performance	–	–	–																
Moderating Variables																			
5 Year of data collection	.138	-.269	-.160	-.049															
6 Degree of customer orientation	-.052	.106	.145	-.014	-.428														
7 Market size	-.042	.086	.011	-.043	-.073	-.057													
8 Degree of social trust	-.054	-.086	.029	-.081	.352	.026	-.467												
9 Alliance network	.095	.345	.156	.283	-.307	.116	.300	-.171											
10 Supplier–customer network	.097	.086	-.015	-.067	.256	-.069	.159	-.076	-.198										
11 Innovation (vs. financial) performance	.272	.310	.134	.180	-.171	.017	.120	-.147	.550	.129									
Control Variables																			
12 Network density dummy	.014	.126	-.064	-.070	-.455	.143	.059	-.136	.275	.031	.310								
13 Network diversity dummy	-.044	–	–	-.067	-.162	.124	.114	-.058	.271	-.054	.100	.207							
14 Objective performance measure	-.128	-.113	-.121	-.262	-.044	.010	.067	-.098	-.120	.033	-.095	.032	.015						
15 Data source	.445	.113	.121	.230	.300	-.066	-.108	.048	.227	.043	.275	-.061	-.028	-.484					
16 Panel data	-.323	-.203	-.023	-.093	-.380	.140	.065	.025	-.023	-.194	-.085	.105	.019	.207	-.526				
17 Partial correlation	-.075	-.229	-.326	-.313	.085	-.056	.108	.021	-.110	.091	-.013	.230	.107	.079	-.005	.058			
18 Published study	.092	.202	.072	.118	-.002	-.291	.507	-.304	.290	.294	.389	.152	.138	-.085	.156	-.169	.006		
19 Top tier journal	.082	.324	.096	.100	-.495	.279	.147	-.150	.412	.006	.435	.462	.337	-.016	.154	-.032	-.022	.382	
20 Western economy	-.176	.017	.102	-.048	-.156	.551	.268	.299	.146	-.129	-.094	.088	.118	-.084	-.113	.122	.046	-.194	.129

All the values in bold are significant at $p < .05$

Table 5 Descriptive statistics and effects of network centrality on firm performance

Examined Relationships	Number of Raw Effects	Number of Studies	Simple Average r	Average Reliability-Adjusted r	Weighted Average Random-Effect r	95% Confidence Interval	Q	I^2 (%)
Panel A (Overall Sample)								
Degree Centrality $\rightarrow$ Firm Performance	628	91	.084	.084	.087	***	[.070, .105]	1.2E+05 ***
Closeness Centrality $\rightarrow$ Firm Performance	331	49	.063	.063	.057	***	***	3.6E+04 ***
Betweenness Centrality $\rightarrow$ Firm Performance	351	52	.049	.049	.050	***	***	2.4E+04 ***
Eigenvector Centrality $\rightarrow$ Firm Performance	389	57	.071	.071	.073	***	***	4.8E+04 ***
Panel B (Network Type: Board Interlock Network)								
Degree Centrality $\rightarrow$ Firm Performance	306	43	.038	.038	.034	***	***	7.3E+04 ***
Closeness Centrality $\rightarrow$ Firm Performance	221	27	.026	.026	.027	***	***	2.6E+04 ***
Betweenness Centrality $\rightarrow$ Firm Performance	228	28	.034	.034	.037	***	***	1.9E+04 ***
Eigenvector Centrality $\rightarrow$ Firm Performance	259	26	.048	.048	.048	***	***	3.2E+04 ***
Panel C (Network Type: Alliance Network)								
Degree Centrality $\rightarrow$ Firm Performance	225	37	.144	.144	.154	***	***	4.5E+04 ***
Closeness Centrality $\rightarrow$ Firm Performance	96	19	.144	.144	.137	***	***	1.6E+03 ***
Betweenness Centrality $\rightarrow$ Firm Performance	93	17	.088	.088	.089	***	***	2.3E+03 ***
Eigenvector Centrality $\rightarrow$ Firm Performance	104	26	.139	.139	.151	***	***	1.3E+04 ***
Panel D (Network Type: Supplier–Customer Network)								
Degree Centrality $\rightarrow$ Firm Performance	97	12	.088	.088	.107	***	***	5.2E+03 ***
Closeness Centrality $\rightarrow$ Firm Performance	14	3	.081	.081	.123	***	***	3.5E+02 ***
Betweenness Centrality $\rightarrow$ Firm Performance	30	7	.036	.036	.040	***	***	2.0E+03 ***
Eigenvector Centrality $\rightarrow$ Firm Performance	26	6	.029	.029	.024	***	***	1.3E+03 ***
Panel E (Performance Measure: Innovation Outcomes)								
Degree Centrality $\rightarrow$ Innovation Outcomes	230	32	.160	.160	.174	***	***	2.9E+04 ***
Closeness Centrality $\rightarrow$ Innovation Outcomes	56	14	.168	.168	.160	***	***	6.0E+02 ***
Betweenness Centrality $\rightarrow$ Innovation Outcomes	75	16	.086	.087	.086	***	***	2.5E+03 ***
Eigenvector Centrality $\rightarrow$ Innovation Outcomes	93	20	.121	.122	.117	***	***	9.9E+03 ***
Panel F (Performance Measure: Financial Outcomes)								
Degree Centrality $\rightarrow$ Financial Outcomes	398	69	.040	.040	.038	***	***	8.7E+04 ***
Closeness Centrality $\rightarrow$ Financial Outcomes	275	41	.041	.041	.039	***	***	3.6E+04 ***
Betweenness Centrality $\rightarrow$ Financial Outcomes	276	43	.038	.038	.040	***	***	2.1E+04 ***
Eigenvector Centrality $\rightarrow$ Financial Outcomes	296	45	.055	.055	.059	***	***	3.7E+04 ***

* Significant at 10% level, two-sided

** Significant at 5% level, two-sided

*** Significant at 1% level, two-sided

The table reports back-transformed Fisher's Z correlations; Q = Cochran's homogeneity test statistic; I² = scale-free index of heterogeneity

Table 6 Effects of moderating factors on the network centrality–firm performance relationship

Panel A (Overall Sample)				
	Model 1 Degree Centrality → Firm Performance	Model 2 Closeness Centrality → Firm Performance	Model 3 Betweenness Centrality → Firm Performance	Model 4 Eigenvector Centrality → Firm Performance
	Coefficient	Coefficient	Coefficient	Coefficient
Year of data collection	-.006 ***	-.007 ***	-.008 ***	.004 *
Degree of customer orientation	.039 ***	.019 ***	-.006 ***	-.028 ***
Market size	.062 ***	.020 **	.009 **	-.021 **
Degree of social trust	.087 ***	.064 ***	.042 ***	-.080 ***
Alliance network	-.063 ***	-.032 ***	-.033 ***	.077 ***
Supplier–customer network	-.002 ***	-.026 ***	.052 ***	-.078 ***
Innovation (vs. financial) performance	.078 ***	.067 ***	.038 ***	.043 **
Network density dummy	.076 ***	-.005 ***	-.077 ***	-.149 ***
Network diversity dummy	.099 ***	—	—	-.139 ***
Objective performance measure	.021 ***	-.055 ***	-.205 ***	-.247 ***
Data source	.497 ***	—	—	-.041 ***
Panel data	-.046 ***	-.201 ***	.021 ***	.004 ***
Partial correlation	-.069 ***	-.025 ***	-.081 ***	-.085 ***
Published study	-.043 **	-.002 **	.016 **	-.007 **
Top tier journal	-.142 ***	.051 ***	-.002 ***	.030 ***
Western economy	-.112 ***	-.051 ***	.027 ***	.055 ***
Constant	11.690 ***	13.780 ***	15.160 ***	-6.656 ***
Number of observations	553	305	344	346

Table 6 (continued)

Panel B (Network Type: Board Interlock Network)				
	Model 1 Degree Centrality → Firm Performance	Model 2 Closeness Centrality → Firm Performance	Model 3 Betweenness Centrality → Firm Performance	Model 4 Eigenvector Centrality → Firm Performance
	Coefficient	Coefficient	Coefficient	Coefficient
Year of data collection	-.002		-.005**	.002
Degree of customer orientation	.025	-.023	.001	-.044*
Market size	.079***	.011	.010	-.024*
Degree of social trust	.071**	.015	.046**	-.080***
Innovation (vs. financial) performance	-.031	.029	.009	.034
Network density dummy	.078*	-.073	.013	—
Network diversity dummy	—	—	—	—
Objective performance measure	-.579***	—	—	—
Data source	-.108	—	—	—
Panel data	-.026	-.099**	.041	.024
Partial correlation	-.029	-.023*	-.074***	-.070***
Published study	-.111***	-.012	.025	-.012
Top tier journal	.008	-.056	-.048	.077
Western economy	-.125***	.005	.006	.061
Constant	2.977	9.862	10.560**	-4.042
Number of observations	304	219	226	253

Table 6 (continued)

Panel C (Network Type: Alliance Network)				
	Model 1 Degree Centrality → Firm Performance Coefficient	Model 2 Closeness Centrality → Firm Performance Coefficient	Model 3 Betweenness Centrality → Firm Performance Coefficient	Model 4 Eigenvector Centrality → Firm Performance Coefficient
Year of data collection	.011 ***	-.019 ***	-.032 ***	.005
Degree of customer orientation	.274 ***	.254 ***	-.41.100 ***	.018
Market size	.269 ***	.014	.005	.060
Degree of social trust	-.245 *	-.276 *	2.661 ***	1.624 ***
Innovation (vs. financial) performance	.112 ***	.075 ***	.093 ***	.122 *
Network density dummy	.099 ***	.145 **	-.075 ***	-.108
Network diversity dummy	-.173 ***	—	—	-.122
Objective performance measure	-.756 ***	-.1.164 ***	21.680 ***	.753
Data source	-.063	—	—	1.025 *
Panel data	-.137 **	—	—	1.314 **
Partial correlation	-.090 ***	-.203 ***	-.176 ***	-.143 ***
Published study	.085 ***	-.259 ***	-.193 ***	-.033
Top tier journal	.096 ***	.146 ***	.090 ***	.000
Western economy	-.332 ***	-.552 ***	65.610 ***	.059
Constant	-22.580 ***	38.430 ***	202.300 ***	-19.660 ***
Number of observations	152	72	88	67

Table 6 (continued)

Panel D (Network Type: Supplier–Customer Network)				
	Model 1 Degree Centrality → Firm Performance Coefficient	Model 2 Closeness Centrality → Firm Performance Coefficient	Model 3 Betweenness Centrality → Firm Performance Coefficient	Model 4 Eigenvector Centrality → Firm Performance Coefficient
Year of data collection	.000	-.659***	-5.909***	.089
Degree of customer orientation	-.317	-2.526***	-131.200***	.082
Market size	-.709	—	-87.250***	-13.510
Degree of social trust	.725	—	279.300***	4.063
Innovation (vs. financial) performance	-.081	—	—	-.132
Network density dummy	.125	—	—	—
Network diversity dummy	—	—	—	—
Objective performance measure	—	—	—	—
Data source	.153	—	—	—
Panel data	-.048	—	21.070***	—
Partial correlation	-.076**	-.014	-.208**	-.071
Published study	-.116	—	—	—
Top tier journal	-.703	—	—	.592
Western economy	.316*	—	86.010***	—
Constant	2.994	1337.300***	11828.800***	-105.100*
Number of observations	97	14	30	26

Table 6 (continued)

Panel E (Performance Measure: Innovation Outcomes)				
	Model 1 Degree Centrality → Innovation Outcomes Coefficient	Model 2 Closeness Centrality → Innovation Outcomes Coefficient	Model 3 Betweenness Centrality → Innovation Outcomes Coefficient	Model 4 Eigenvector Centrality → Innovation Outcomes Coefficient
Year of data collection	-.021 ***	-.021 **	-.010 ***	.000
Degree of customer orientation	.252 ***	-.304	.006	-.144
Market size	.273 ***	-.703 *	.109	-.357
Degree of social trust	-.106 ***	-1.039 **	-.020	-.190
Alliance network	.006	-.354	-.107	-.022
Supplier–customer network	.093 **	–	–	-.287 ***
Network density dummy	.094 *	.004	-.102 ***	-.013
Network diversity dummy	-.011	–	–	-.186 **
Objective performance measure	.035	-.855 ***	-.454 **	-.196
Data source	.675 ***	–	–	-.180
Panel data	-.336 ***	–	–	–
Partial correlation	-.172 ***	-.132 *	-.122 ***	-.019
Published study	.239 ***	.960 *	-.030	.529
Top tier journal	-.398 ***	.210 ***	.011	-.048
Western economy	-.368 ***	.498	.043	.383 **
Constant	39.920 ***	52.500 **	20.330 ***	2.708
Number of observations	176 47		71	59

Table 6 (continued)

Panel F (Performance Measure: Financial Outcomes)				
	Model 1 Degree Centrality → Financial Outcomes	Model 2 Closeness Centrality → Financial Outcomes	Model 3 Betweenness Centrality → Financial Outcomes	Model 4 Eigenvector Centrality → Financial Outcomes
	Coefficient	Coefficient	Coefficient	Coefficient
Year of data collection	-.003	-.007***	-.008***	.002
Degree of customer orientation	.032*	.017	.013	-.041*
Market size	.068***	.022**	.015	-.017
Degree of social trust	.072***	.065***	.060***	-.073**
Alliance network	-.087***	-.009	-.027	.071**
Supplier–customer network	-.057**	-.029	.044**	-.017
Network density dummy	.068	-.011	.020	-.217***
Network diversity dummy	.067	–	–	–
Objective performance measure	-.778***	–	–	-.406***
Data source	-.039	-.198***	.011	.007
Panel data	-.029**	-.030**	-.077***	-.095***
Partial correlation	-.084***	-.002	.023	-.021
Published study	-.006	.013	-.054	.021
Top tier journal	-.116***	-.052**	-.002	.054*
Western economy	–	–	–	-.213**
Constant	5.933	13.130***	14.970***	-3.208
Number of observations	377	258	273	287

* Significant at 10% level, two-sided

** Significant at 5% level, two-sided

*** Significant at 1% level, two-sided

4: $\alpha = 0.077$, $p < 0.01$). The quality and richness of resources that more influential strategic partners hold in alliance networks underscore the pivotal role of eigenvector centrality in fostering collaborative advantages and resource access within such networks.

Also, betweenness centrality holds greater significance in supplier–customer networks when contrasted with board interlocks (Model 3: $\alpha = 0.052$, $p < 0.05$). Serving as a gatekeeper to regulate the flow of information and resources within a supply network can furnish a firm with enhanced bargaining leverage. This aspect becomes particularly crucial in supply networks established to bolster efficiency and agility. On the contrary, the association between eigenvector centrality and firm performance exhibits less strength in supply networks in comparison to board interlocks (Model 4: $\alpha = -0.078$, $p < 0.01$). This could be explained by the presence of influential peers within a firm's supply network. When a firm is associated with such peers, it may be in a situation of diminished bargaining power, hindering its ability to capitalize on lower sourcing costs or wield stronger pricing power.

Finally, a central position consistently shows a stronger association with innovativeness compared to financial outcomes (Model 1: $\alpha = 0.078$, $p < 0.01$; Model 2: $\alpha = 0.067$, $p < 0.01$; Model 3: $\alpha = 0.038$, $p < 0.1$; Model 4: $\alpha = 0.043$, $p < 0.1$). This observed pattern remains consistent regardless of the type of centrality measure utilized. High centrality facilitates access to a greater volume of high-quality information that can foster the realization of innovation opportunities (Fang et al., 2016). Yet, not all innovations generate economic returns, and a central position can be expensive for firms. For example, ties that position a firm more deeply in a network can increase coordination costs (Ahuja, 2000). Therefore, it is reasonable to expect the association of centrality with financial outcomes to be weaker than its relationship with innovativeness.

Complementary analyses

As we indicated, substantial heterogeneity exists in previous studies concerning the network type and firm performance measures they use, which could influence the moderating effects of contingency factors. We conduct a series of supplementary analyses to investigate the impact of methodological characteristics on the moderating effects of contextual factors. In detail, we break down our sample according to network type or performance measure, then estimate Eqs. 2–5 for each subsample. We provide the results in Table 6, Panels B–F. We find that the moderating effects of contextual factors on the relationship between network centrality and firm performance depend, at least

partially, on the type of network and how firm performance is measured.

For instance, the impact of degree centrality on firm performance within alliance networks has shown a strengthening trend over time (Panel C, Model 1: $\alpha = 0.011$, $p < 0.01$). Degree centrality enables firms to pool resources with numerous partners. This advantage has become increasingly vital in alliance networks, as keeping pace with rapidly evolving industry developments poses significant challenges to individual firms. Pursuing innovation, for example, has become more demanding with the accelerated evolution of technologies, necessitating larger investments to address the complexities of R&D initiatives. Maintaining a higher degree centrality within alliance networks can help firms alleviate their financial burdens by leveraging resources from other strategic partners.

Robustness checks

To check the robustness of our findings, we first assess the possibility of publication bias, or the file drawer problem, by calculating Rosenthal's (1979) fail-safe sample size. The fail-safe sample sizes for the significant results exceed 25,341, suggesting that the observed significant effects are not susceptible to a file-drawer problem. In addition, before estimating Eqs. 2–5, we checked for potential multicollinearity among the independent and control variables. Web Appendix B shows that across all models, the variance inflation factors obtained from regressing the Fisher's Z coefficients on the independent variables are less than the suggested threshold of 10 (Hair et al., 2010). This finding mitigates concerns about multicollinearity in our models.

Finally, we estimated Eqs. 2–5 without imputing the missing values of independent variables. Opting not to impute the missing values reduces the sample size to 705 observations. As Web Appendix C details, our findings remain mostly unchanged under this alternative specification. For example, the associations of closeness and betweenness centralities with performance have weakened over time. The degree of customer orientation, market size, and degree of social trust positively influence the network centrality–performance relationship, mirroring the findings from the main analyses conducted with the full sample. An exception concerns the effects of the year of data collection in Models 1 and 4, as they are no longer statistically significant. This is likely because the non-imputation of missing values constrains the range of values for this variable from 1987–2020 to 2006–2017, which reduces its variation (as indicated by its standard deviation) by more than 50%. This reduction, in turn, likely limits the variable's ability to adequately capture the variations in the dependent variables.

Discussion

With a meta-analysis of extant literature related to organizational networks, we establish that, on average, network centrality is positively associated with firm performance. Interestingly, the overall performance effects of degree, closeness, betweenness, and eigenvector centralities exhibit variations in magnitude, and contingency factors moderate the relationships between these centrality measures and firm performance differently. Our findings indicate that the network centrality–firm performance relationship is dynamic and has evolved over time. It also depends on the extent of customer orientation in the wider economy, market size, and the degree of social trust. Moreover, our analyses reveal that the network centrality–performance relationship and the moderating effects of contextual factors depend, at least partially, on the network type (i.e., board interlock vs. alliance vs. supplier–customer) as well as the measure of firm performance (i.e., innovation vs. financial outcomes).

When considering the results collectively, we observe intriguing patterns regarding the centrality–performance relationship. For example, we note that the associations of closeness and betweenness centralities with firm performance have decreased over time, irrespective of network type or the metric used to gauge firm performance (see Table 6). Closeness represents a firm’s ability to access information more quickly than peripheral firms, whereas betweenness centrality denotes a firm’s potential to function as an information gatekeeper. Our results suggest that these positional advantages have likely diminished over time due to recent technological advances, which have democratized information access even for less connected firms. Similarly, regardless of how centrality is measured, it has a stronger relationship with innovativeness compared to financial outcomes in the overall sample (see Table 6, Panel A). Centrality provides access to abundant high-quality information, nurturing innovation prospects (Fang et al., 2016). However, not all innovations yield economic returns, and maintaining centrality can be costly for firms.

Theoretical contributions

This research makes several contributions to marketing literature pertaining to organizational networks. Specifically, previous studies in marketing frequently explore the performance effects of different centrality measures, including degree, closeness, betweenness, and eigenvector centralities (see, e.g., Fang et al., 2016; Srinivasan et al., 2018). Our research adds to this literature by highlighting that

the overall net performance impacts of various centrality measures are different in magnitude, and these effects display notable variations in different temporal and competitive conditions. With this finding, we caution researchers against aggregating different centrality measures, such as in a component factor analysis, in their effort to simplify the interpretation of performance effects of network centrality.

For example, Larcker et al. (2013) employ factor analysis to construct a composite measure that consolidate various centrality measures into a proxy for the overall connectedness of a firm within a network. However, our findings underscore the importance of recognizing that each centrality measure captures unique facets of a firm’s network connectivity. Degree centrality measures direct connections; closeness centrality emphasizes proximity within a network; betweenness centrality reflects a firm’s bridging role; and eigenvector centrality underscores the significance of well-connected firms in a network. Each measure offers a distinct perspective on the nature and significance of inter-firm relationships. Consequently, consolidating these measures can obscure subtle, pivotal nuances in the performance implications of a firm’s centrality within its network.

In addition, our findings contribute to a more nuanced view of the connection between network centrality and firm performance because we integrate contextual factors into social network analysis, which is in line with the perspective advocated by Shipilov (2006). Specifically, this study is among the first to document the dynamic nature of network centrality–firm performance linkages. Future studies should use more comprehensive and longer-duration data sets, which are better equipped to capture the temporal dynamics inherent in this relationship.

Previous research has primarily focused on the performance implications of network centrality from a structural standpoint. Our study complements this perspective by emphasizing the role of contextual factors in shaping the positional advantages of network centrality. Specifically, we observe a stronger link between holding a central position and firm performance in more customer-oriented economies, larger markets, and environments with higher social trust. We also highlight the role of methodological factors such as the type of network and firm performance measure in determining the effectiveness of network centrality. These findings shed light on the context-dependent nature of the centrality–performance relationship and emphasize the imperative to consider broader environmental and methodological conditions when assessing the effectiveness of network centrality. Researchers should incorporate such factors into their organizational network analyses to fully grasp network centrality’s performance implications and potential.

Managerial implications

Noting the context-dependent nature of the relationship between network centrality and firm performance, we strongly encourage firms to identify and continuously evaluate the network structure of the industry in which they operate, along with their position within that network. This point directly informs senior executives that occupying a central position does not necessarily guarantee better firm performance. Our findings can also help managers identify the boundary conditions for the effectiveness of centrality when developing their networking strategies. For example, the informational advantages of central firms become increasingly valuable in highly customer-oriented markets, as the market intelligence acquired through network centrality empowers firms to identify gaps in customer needs and requirements more promptly. However, firms occupying central positions in customer-oriented markets must exercise caution regarding its potential financial performance implications. Addressing customers' unique needs and requirements can impose substantial costs on firms, potentially eroding the positive impacts of network centrality.

Additionally, to succeed in larger markets marked by the widespread dispersion of information, firms must strategically position themselves at the core of their pertinent networks. This endeavor involves actively seeking collaboration opportunities with industry stakeholders and forming partnerships—initiatives that can enhance their visibility within networks and establish them as indispensable contributors to the network's overall dynamics.

The strategic interests of network members do not always align, which can create conflicts of interest and power struggles in the network and prevent effective coordination and information sharing (see Gulati, 1998). Firms with higher centrality should promote mutual trust to foster cooperation and communication. Open and transparent communication, an atmosphere where information flows freely among network participants, active pursuits of mutually beneficial outcomes, professional conflict resolution, adherence to ethical standards, sharing both risks and rewards, and nurturing personal relationships are all essential elements that can collectively contribute to the establishment and strengthening of trust within a network.

Because occupying a central position in various types of networks can yield different performance outcomes, firms should carefully distinguish among the networks they are a part of. Lastly, firms should adjust their networking strategies to match their performance objectives. Centrality typically exhibits a more pronounced association with firm innovation because it provides access to an extensive pool of valuable information that can stimulate innovative opportunities (Fang et al., 2016). Yet connections that embed a firm more deeply within a network can also increase the costs

associated with efficiently managing relationships with other network members (Ahuja, 2000). We offer a summary of the theoretical and managerial implications of this meta-analysis in Table 7.

Toward a research agenda

Our findings suggest several important avenues for further research. Although not exhaustive, the following discussion highlights some key issues that demand attention to advance the use of social network analysis in marketing.

Exploring the dynamic and transitional nature of organizational networks

Organizational networks are based on interfirm relationships, and “relationships are dynamic in nature” (Zhang et al., 2016, p. 53). Therefore, organizational networks' characteristics and functions will evolve. Our study documents the dynamic nature of the network centrality–performance relationship over time. Continued research endeavors could explore the temporal dynamics inherent in the performance effects of other dimensions of network position.

For example, in Web Appendix D, we present the findings regarding the temporal trends in the relationship between network brokerage and firm performance. Network brokerage represents the extent to which a firm bridges structural gaps and connects otherwise disparate nodes (Burt, 1992). We find that even as the relationship between network brokerage and innovativeness has strengthened over time (Model 1: $\alpha = 0.009$, $p < 0.05$), the link between brokerage and financial performance has waned (Model 2: $\alpha = -0.004$, $p < 0.01$). This discovery underscores a noteworthy trend in organizational networks that we posit is likely a consequence of technological advancements. With the proliferation of communication technologies, firms have become increasingly interconnected, resulting in denser networks. This, in turn, demands greater allocations of resources, including finances and time, to sustain a brokerage position and maintain a competitive edge within the network. As such, the effectiveness of brokerage for improving financial performance has diminished over time.

Beyond temporal dynamics, studies could analyze structural data to identify different stages of a network's lifecycle (e.g., initiation, growth, maturity, decline, renewal), with the recognition that changes and transitions in a network do not necessarily happen in a linear or chronological sequence. Stages in a network lifecycle may unfold unpredictably or irregularly, influenced by various factors that are not directly related to time. To expand our results, additional research might investigate how a network's transitions over its lifecycle phases affect the resources and capabilities (e.g., financial, reputational, technological

Table 7 Implications for theory and practice

Key Findings	Theoretical Implications	Managerial Implications
On average, network centrality is positively associated with firm performance. However, there is substantial heterogeneity across previous studies in terms of the direction and magnitude of the centrality–performance relationship.	The current study advances social network theory by offering a comprehensive synthesis across existing studies regarding the performance effects of network centrality.	Firms need to understand the network structure of the industry in which they operate, as well as their position within that network. A central position within an organizational network does not necessarily guarantee stronger firm performance.
The overall performance effects of various centrality measures are different in magnitude, and their associations with firm performance are contingent on environmental factors, network characteristics, and the specific performance measures employed.	We caution researchers against overlooking the distinctions among different centrality measures and advise against employing composite measures, such as one that attempts to capture a firm's overall connectedness using various indicators of network centrality (e.g., Larcker et al., 2013). Such an approach could hinder researchers from accurately identifying the true impact of network position on firm performance in varying conditions. Also, researchers should diligently consider the potential interactions between different centrality measures and contextual factors.	Different centrality measures yield distinct impacts on firm performance, according to varying contextual boundaries. It is imperative for managers to be cognizant of these conditions when making decisions about which centrality measures to emphasize. For example, higher degree or closeness centralities prove more advantageous for firms operating in larger economies marked by the widespread dispersion of information.
The associations of degree, closeness, and betweenness centralities with firm performance have diminished over time. Meanwhile, the eigenvector centrality–performance relationship has strengthened.	Research must account for time dynamics when evaluating the effectiveness of a central position within a network.	Managers should recognize that the relationship between network centrality and firm performance is dynamic and can change over time. Therefore, they should regularly assess and calibrate their networking strategies to stay competitive.
The relationship between network position and firm performance depends on (a) the extent of customer orientation, (b) market size, and (c) the degree of social trust	Contextual and methodological factors should be incorporated into analyses of the performance effects of network centrality. Failing to do so may limit the generalizability of empirical findings.	In customer-oriented economies, where anticipating emerging trends and addressing customer needs to enhance satisfaction is pivotal for remaining competitive, establishing a central position can significantly enhance a firm's market-sensing capabilities. In turn, these capabilities enable more effective tracking and responsive adaptation to the ever-evolving needs and preferences of customers
Also, the performance effects of network centrality and the moderating effects of the aforementioned contextual factors depend, at least partially, on network type (i.e., board interlock vs. alliance vs. supplier–customer) and the performance measure (i.e., innovation vs. financial outcomes).		Network members should cultivate trust among themselves to dismantle barriers to information exchange, thereby enabling access to detailed, sensitive, or proprietary information that might otherwise remain confidential or undisclosed Managers should carefully consider both the nature of the network and their specific performance objectives when formulating networking strategies.

knowledge, market sensemaking) required to secure a central position. Another line of studies could explore whether and how holding a superior network position affects different performance dimensions in varying states of network evolution. Exploring the moderating effects of contextual factors in each phase of a network's lifecycle may also be fruitful. For example, during the initial phase, often characterized by loosely connected structures, information exchanges may occur over longer pathways and take longer to complete. In such circumstances, accessing communication technologies that facilitate information sharing may help central firms benefit more from their positional advantages.

Mediating mechanisms of the network centrality–performance linkage

Previous studies primarily focus on the performance outcomes of network centrality; still, considerations of the mechanisms by which network centrality affects firm performance remain largely conceptual and need to be validated empirically. Organizational network literature conceptualizes the role of several mechanisms, such as information reach and richness, power and dependency, and conflicts of interest and opportunism, in linking centrality to firm performance. Further studies could document such mediating effects empirically and specify each mechanism's relative impact and boundary conditions. Also, research is needed to explore the state dependence of these mechanisms across the lifecycle of a network. For example, trust may dominate the initiation phase as a mediating mechanism, whereas opportunism could become the key mediator in the decline stage.

Conclusion

In conclusion, this study synthesizes the existing literature on the performance implications of network centrality within organizational networks. By doing so, we address a notable research emphasis for marketing scholars, urging the adoption of a social network perspective to elucidate marketing matters (Deighton et al., 2021). Our findings highlight substantial heterogeneity across prior studies regarding the performance effects of network centrality, mainly attributed to variations in centrality measurement, contextual conditions (i.e., temporal dynamics, degree of customer orientation, market size, and social trust levels), and methodological factors (i.e., network types and performance metrics). We encourage future studies to integrate such factors into their analyses to bolster the generalizability of their findings.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11747-024-01043-8>.

Declarations

Conflicts of interest The authors declare that they have no conflicts of interest.

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